



## IV SEMESTER B.TECH. (COMPUTER SCIENCE & ENGINEERING) END SEMESTER MAKEUP EXAMINATION, MAY 2023

### SUBJECT: OPERATING SYSTEMS [CSE 2272] REVISED CREDIT SYSTEM

Time: 3 Hours

MAX. MARKS: 50M

#### Instructions to Candidates:

- ❖ Answer **ALL** the questions.
- ❖ Missing data may be suitably assumed.

- 1A.** Write bankers algorithm for dead lock avoidance. Considering a system with five processes P0 through P4 and three resource types A, B, C is given below. i) Calculate the need matrix. ii) Check whether the system is safe or not? If safe, what is the safe sequence? iii) Calculate the total sum of each type of resource?

(5M)  
(CO3,  
L2,L3)

| Processes | Allocation |   |   | Max |   |   | Available |   |   |
|-----------|------------|---|---|-----|---|---|-----------|---|---|
|           | A          | B | C | A   | B | C | A         | B | C |
| P0        | 1          | 1 | 2 | 4   | 3 | 3 | 2         | 1 | 0 |
| P1        | 2          | 1 | 2 | 3   | 2 | 2 |           |   |   |
| P2        | 4          | 0 | 1 | 9   | 0 | 2 |           |   |   |
| P3        | 0          | 2 | 0 | 7   | 5 | 3 |           |   |   |
| P4        | 1          | 1 | 2 | 1   | 1 | 2 |           |   |   |

Table Q.1A

- 1B.** Describe how the Swap() instruction can be used to provide mutual exclusion that satisfies the bounded waiting requirement.

(3M)  
(CO2, L2 )

- 1C.** Consider process P<sub>0</sub> is interrupted during its execution and there is a context-switch to another Process P<sub>1</sub> by the CPU. Describe the action taken by a kernel to context-switch between the processes with a diagram.

(2M)  
(CO1, L2 )

- 2A.** Explain segmentation architecture. How is protection achieved in segmentation? Consider five memory partitions of size 100 KB, 500 KB, 200 KB, 450 KB and 600 KB in same order. If sequence of requests for blocks of size 212 KB, 417 KB, 112 KB and 426 KB in same order come, Apply First fit, worst fit and best fit algorithms for memory allocations. Find the optimal algorithm for memory allocation and justify.

(5M)  
(CO4,  
L2,L3)

| <b>2B.</b> | <p>i. List the roles of Lazy swapper.</p> <p>ii. Calculate Average waiting time and Average turnaround time for non-preemptive SJF and preemptive SJF with the help of a Gantt Chart for the following data.</p> <table border="1"> <thead> <tr> <th>Process</th><th>Arrival Time (msec)</th><th>Burst Time(msec)</th></tr> </thead> <tbody> <tr> <td>P1</td><td>0</td><td>8</td></tr> <tr> <td>P2</td><td>4</td><td>5</td></tr> <tr> <td>P3</td><td>3</td><td>2</td></tr> <tr> <td>P4</td><td>2</td><td>1</td></tr> </tbody> </table> <p>Table Q.2B</p> | Process            | Arrival Time (msec) | Burst Time(msec) | P1 | 0 | 8 | P2 | 4 | 5 | P3 | 3 | 2 | P4 | 2 | 1 | (3M)<br>(CO4,<br>CO3)(L2,L3) |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|---------------------|------------------|----|---|---|----|---|---|----|---|---|----|---|---|------------------------------|
| Process    | Arrival Time (msec)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Burst Time(msec)   |                     |                  |    |   |   |    |   |   |    |   |   |    |   |   |                              |
| P1         | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 8                  |                     |                  |    |   |   |    |   |   |    |   |   |    |   |   |                              |
| P2         | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 5                  |                     |                  |    |   |   |    |   |   |    |   |   |    |   |   |                              |
| P3         | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2                  |                     |                  |    |   |   |    |   |   |    |   |   |    |   |   |                              |
| P4         | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1                  |                     |                  |    |   |   |    |   |   |    |   |   |    |   |   |                              |
| <b>2C.</b> | Explain the microkernel operating system structure with diagram, benefits, and detriments.                                                                                                                                                                                                                                                                                                                                                                                                                                                               | (2M)<br>(CO1, L2 ) |                     |                  |    |   |   |    |   |   |    |   |   |    |   |   |                              |
| <b>3A</b>  | Apply FIFO, LRU and optimal page replacement algorithms for the following page reference string. Page reference string 7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1. Assume 3 frames and all frames are initially empty. Calculate the number of page faults for each algorithm.                                                                                                                                                                                                                                                                              | (5M)<br>(CO4,L3)   |                     |                  |    |   |   |    |   |   |    |   |   |    |   |   |                              |
| <b>3B</b>  | Explain the handling of a user application invoking the open () system call with a neat diagram and state the purpose of medium-term scheduler with neat diagram.                                                                                                                                                                                                                                                                                                                                                                                        | (3M)<br>(CO1,L2)   |                     |                  |    |   |   |    |   |   |    |   |   |    |   |   |                              |
| <b>3C</b>  | Explain in detail about threading and its models with neat diagram. Explain the challenges in multicore programming.                                                                                                                                                                                                                                                                                                                                                                                                                                     | (2M)<br>(CO1,L2)   |                     |                  |    |   |   |    |   |   |    |   |   |    |   |   |                              |
| <b>4A</b>  | Explain in detail about page table structure and its variation with suitable example. Define fragmentation and suggest the solution to reduce external fragmentation.                                                                                                                                                                                                                                                                                                                                                                                    | (5M)<br>(CO4,L2)   |                     |                  |    |   |   |    |   |   |    |   |   |    |   |   |                              |
| <b>4B</b>  | Let the page fault service time be 10ms in a computer with average memory access time being 20ns. If one page fault is generated for every $10^6$ memory accesses, what is the effective access time for the memory? List steps involved in page fault service.                                                                                                                                                                                                                                                                                          | (3M)<br>(CO4,L3)   |                     |                  |    |   |   |    |   |   |    |   |   |    |   |   |                              |
| <b>4C</b>  | Explain sequential and direct access methods of a file with examples.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | (2M)<br>(CO5,L2)   |                     |                  |    |   |   |    |   |   |    |   |   |    |   |   |                              |
| <b>5A</b>  | Assume that a disk drive has cylinders numbered 0 to 4999. Cylinder 143 is currently serving by drive, and the cylinder 125 was a previous request. The 86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130 are pending requests. Suggest any two best disk-scheduling algorithms from FCFS, SSTF, SCAN, C-SCAN and C-LOOK, those uses minimum distance to satisfy all the pending requests. Justify your answer.                                                                                                                                            | (5M)<br>(CO5,L3)   |                     |                  |    |   |   |    |   |   |    |   |   |    |   |   |                              |
| <b>5B</b>  | Explain the any three ways of implementing the access matrix.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | (3M)<br>(CO5,L2)   |                     |                  |    |   |   |    |   |   |    |   |   |    |   |   |                              |
| <b>5C</b>  | Discuss the strengths and weaknesses of implementing an access matrix using capabilities that are associated with domains.                                                                                                                                                                                                                                                                                                                                                                                                                               | (2M )<br>(CO5,L2)  |                     |                  |    |   |   |    |   |   |    |   |   |    |   |   |                              |

# Question Paper

Exam Date & Time: 26-Jun-2024 (02:30 PM - 05:30 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

FOURTH SEMESTER B.TECH END SEMESTER MAKE-UP EXAMINATIONS, JUNE 2024

**OPERATING SYSTEMS [CSE 2226]**

**Marks: 50**

**Duration: 180 mins.**

**Answer all the questions.**

Instructions to Candidates: Answer ALL questions Missing data may be suitably assumed

- 1) With the aid of a neat diagram, explain the layered structure of operating system. Identify any three advantages and disadvantages of the layered structure. (5)
- A)
- B) Discuss any four states of a process. Point out any two differences between a process and a program. (3)
- C) Describe the resources that are used when a thread is created. How do they differ from those used when a process is created. (2)

- 2) Analyze the major drawback of priority scheduling algorithms. Describe a solution for the same. (5)
- A) Assume that the following processes are scheduled using shortest remaining time first scheduling algorithm. Use FIFO to break a tie wherever required.

| Process | Arrival Time | Burst Time |
|---------|--------------|------------|
| P1      | 0            | 8          |
| P2      | 1            | 4          |
| P3      | 2            | 2          |
| P4      | 3            | 1          |
| P5      | 4            | 3          |
| P6      | 5            | 2          |

- (a) Show the scheduling order of the processes using Gantt chart.
- (b) Determine the turnaround time and waiting time for each process.

- B) There are 5 resources R1 with 2 instances, R2 with single instance, R3 with 2 instances, R4 with single instance and R5 with single instance (3)

- (a) Process P1 holding one instance of R1 and an instance of R2 and requesting one instance of R5
- (b) Process P2 holding one instance of R1 and an instance of R4.
- (c) Process P3 is holding one instance of R3 and requesting one instance of R2.
- (d) Process P4 is holding one instance of R3 and requesting for one instance of R4 and requesting one instance of resource R5
- (e) Process P5 is requesting for one instance of R3 and holding one instance of R5.
- (f) Process P6 is requesting for one instance of R4.

- Draw the resource allocation graph and identify whether deadlock exists? If yes, how many processes are involved in the deadlock? If No, give reason. Justify your answer depending on Yes/No.
- C) Write the structure of a process  $P_i$  in Peterson's solution. Justify how mutual exclusion is preserved in Peterson's solution. (2)
- 3) (i) Explain the steps involved in basic page replacement with a diagram. (2Marks) (5)
- A) (ii) Consider the following page reference string: 6, 7, 8, 9, 6, 7, 1, 6, 7, 8, 9, 1
- How many page faults would occur for the following replacement algorithms, with three page frames? Remember that all frames are initially empty. Show the frame contents in each step. (3Marks)
- (a) FIFO replacement
- (b) Optimal algorithm
- B) Consider a logical address space of 8 pages; each page is 2048 byte long, mapped onto a physical memory of 64 frames. (3)
- (i) How many bits are there in the logical address and how many bits are there in the physical address?
- (ii) A 6284 bytes program is to be loaded in some of the available frames = {10, 8, 40, 25, 3, 15, 56, 18, 12, 35}. Show the contents of the program's page table.
- (iii) Is there internal fragmentation? If yes what is the size of internal fragmentation?
- C) Describe the cause of thrashing? How can the system prevent thrashing using working-Set model?. (2)
- 4) Paging uses Page table associated with each process. Each entry refers to a page and processes reference pages through the pages' virtual addresses. When there are multiple processes, each Page table consists of millions of entries and consumes large amounts of physical memory. (i) Propose a solution to the above problem where a large amount of physical memory for page tables is required. (5)
- A) (ii) With the help of a labelled diagram show how this could be done for a 32-bit logical address space of 4-KB page size and 10 bit page number and explain the same
- B) Consider a disk drive which has 200 cylinders, numbered 0 to 199. The drive is currently serving a request at cylinder 60. The queue of pending requests, in order, is: 70, 140, 50, 125, 30, 25, 160. Starting from the current head position, calculate what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for following disk-scheduling algorithm? Depict the sequence in a graph. (3)
- (a) SCAN (b) C-SCAN
- C) How does the principle of least privilege aid in the creation of protection system. Justify (2)
- 5) Explain any four file operations. (4)
- A)
- B) Write a short note on (3)
- (a) Two level Directory
- (b) File System Mounting.
- C) Explain "Global table" and "Access list" implementation of access matrix. (3)

-----End-----

# Question Paper

Exam Date & Time: 09-May-2024 (02:30 PM - 05:30 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

FOURTH SEMESTER B.TECH END SEMESTER EXAMINATIONS, MAY 2024

**OPERATING SYSTEMS [CSE 2226]**

**Marks: 50**

**Duration: 180 mins.**

**Answer all the questions.**

Instructions to Candidates: Answer ALL questions Missing data may be suitably assumed

- 1) Explain four benefits of multithreading. Point out two examples in which multithreading provides better performance than a single-threaded solution. Identify and explain the system calls that needs to be executed by a command interpreter or shell in order to start a new process with different code. (5)
- A)
- B) Differentiate the two communication methods with respect to its operation, primitives, and properties of communication link in message passing system associated with naming feature. (3)
- C) Identify the associated queues, required scheduler and the state of process when the process needs to be (2)

(a) Executed by CPU

(b) Loaded into main memory.

- 2) Assume that the following processes are scheduled using round robin scheduling algorithm with a time quantum of 2 milliseconds. (5)

A)

| Process | Arrival Time | Burst Time |
|---------|--------------|------------|
| P2      | 0            | 6          |
| P4      | 0            | 1          |
| P3      | 5            | 3          |
| P6      | 5            | 4          |
| P5      | 6            | 9          |
| P1      | 7            | 15         |

(a) Show the scheduling order of the processes using ready queue and Gantt chart.

(b) Determine the turnaround time and waiting time for each process.

(c) Mention the number of context switches.

B)

For the table given below, use Banker's Algorithm and identify safe sequence

(3)

| Processes | Allocation | Max     | Available |
|-----------|------------|---------|-----------|
|           | A B C D    | A B C D | A B C D   |
| P1        | 0 0 1 2    | 0 0 1 2 | 1 5 2 0   |
| P2        | 1 0 0 0    | 1 7 5 0 |           |
| P3        | 1 3 5 1    | 2 3 5 6 |           |

| P4 | 0 6 3 2 | 0 6 5 2 |  |
|----|---------|---------|--|
| P5 | 0 0 1 4 | 0 6 5 6 |  |

- C) Show how semaphore can be used to control access to a resource with finite number of instances by multiple processes with an example. (2)
- 3) (i) Assume a demand-paged memory. Consider memory access time of 100 nanoseconds. It take 8 milliseconds to service a page fault if an empty page is available or the replaced page is not modified. What is the maximum acceptable page-fault rate for an effective access time of no more than 200 nanoseconds? (2Marks) (5)
- A) (ii) Consider the following page reference string:4, 3, 2, 1, 2, 1, 5, 4, 6, 7, 2, 3, 1, 6, 4, 3, 1, 3, 4, 6. How many page faults would occur for the following replacement algorithms, with three page frames? Remember that all frames are initially empty. Show the frame contents in each step. (3marks)
- (a) LRU replacement
- (b) Optimal algorithm
- B) Consider a page table with Translation look-aside buffer(TLB) added, explain how the TLB will improve the memory access time with a neat labelled diagram. Also, calculate the effective memory reference time for a page table with TLB and 75 percent of all page-table references are found in the TLBs. Assume that finding a page-table entry in the TLBs takes 2 nanoseconds, if the entry is present. (3)
- C) Define stub and explain its role in implementing the dynamic linking feature of Operating system. (2)
- 4) a) Sketch a neat diagram depicting segmentation hardware. Consider the segmentation table given below. (5)

A)

| Segment No. | Base | Limit |
|-------------|------|-------|
| 0           | 1400 | 1000  |
| 1           | 6300 | 400   |
| 2           | 4300 | 400   |
| 3           | 3200 | 1100  |
| 4           | 4700 | 1000  |

Which of the following references will produce trap addressing error? Provide justification for each.

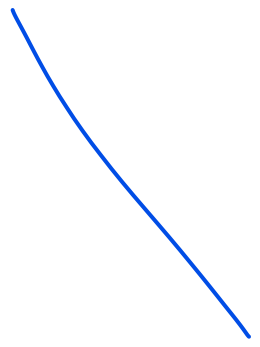
- (i) segment 2, byte 53
- (ii) segment 0, byte 1222

b) Consider

tables? Justify your answer.

- (i) A conventional single-level page table
- (ii) An inverted page table

- B) Suppose that a disk drive has 200 cylinders, numbered 0 to 199. The drive is currently serving a request at cylinder 50. The queue of pending requests, in order, is: 82,170,43,140,24,16,190. Starting from the current head position, calculate what is the seek sequence in a graph. (3)
- C) Consider three domains (D1, D2, D3) and four files (f1,f2, f3, f4). Represent the set of access rights (2)



✓



| 2A.     | Apply FIFO, LRU and optimal page replacement algorithms for the following page reference string: 6, 1, 1, 2, 0, 3, 4, 6, 0, 2, 1, 2, 1, 2, 0, 3, 2, 1, 2, 0. Assume 3 frames, all frames are initially empty. Calculate the number of page faults for each algorithm.                                                                                                                                                                                                                                                                                                                                                                                  | (5M)<br>(CO4, L3)    |              |            |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |                             |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|--------------|------------|----|---|---|----|---|---|----|---|---|----|---|---|----|---|---|----|---|---|-----------------------------|
| 2B.     | <p>i. Scheduling the following processes P1,P2,P3,P4,P5,P6 using Round Robin algorithm. Illustrate the scheduling using Gantt Chart. Calculate Average Waiting time and Turnaround time. Quantum=2ms.</p> <table border="1"> <thead> <tr> <th>Process</th><th>Arrival Time</th><th>Burst Time</th></tr> </thead> <tbody> <tr> <td>P1</td><td>0</td><td>4</td></tr> <tr> <td>P2</td><td>1</td><td>5</td></tr> <tr> <td>P3</td><td>2</td><td>2</td></tr> <tr> <td>P4</td><td>3</td><td>1</td></tr> <tr> <td>P5</td><td>4</td><td>6</td></tr> <tr> <td>P6</td><td>6</td><td>3</td></tr> </tbody> </table> <p>ii. Write a short note on copy on write.</p> | Process              | Arrival Time | Burst Time | P1 | 0 | 4 | P2 | 1 | 5 | P3 | 2 | 2 | P4 | 3 | 1 | P5 | 4 | 6 | P6 | 6 | 3 | (3M)<br>(CO3, CO4, L15, L3) |
| Process | Arrival Time                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Burst Time           |              |            |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |                             |
| P1      | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 4                    |              |            |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |                             |
| P2      | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 5                    |              |            |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |                             |
| P3      | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2                    |              |            |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |                             |
| P4      | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1                    |              |            |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |                             |
| P5      | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 6                    |              |            |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |                             |
| P6      | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 3                    |              |            |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |                             |
| 2C.     | Explain about the inter process communication using shared memory with suitable codesnippet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | (2M)<br>(CO1, L2)    |              |            |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |                             |
| 3A      | Explain in detail about LRU approximation algorithms with suitable example.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | (5M)<br>(CO4,L2)     |              |            |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |                             |
| 3B      | Explain race condition with an example and provide hardware based solution to address the problem in a detailed manner.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | (3M)<br>(CO2,L2)     |              |            |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |                             |
| 3C      | With the help of neat diagram, explain different states of a process. Mention the role of Process Control Block along with all the pieces of information that is stored related to the process with neat diagram.                                                                                                                                                                                                                                                                                                                                                                                                                                      | (2M)<br>(CO1,L2)     |              |            |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |                             |
| 4A      | Explain in detail about fixed and global allocation of frames with an example. Let us assume if there are 100 frames and 5 processes. Calculate frames allocated for process using fixed allocation and proportional allocation.                                                                                                                                                                                                                                                                                                                                                                                                                       | (5M)<br>(CO4,L2,L3)  |              |            |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |                             |
| 4B      | Consider a single level paging scheme with TLB. Assume no page fault occurs. It takes 100 ns to access the physical memory. If TLB hit ratio is 60% and effective memory access time is 160 ns, find TLB access time. Define thrashing.                                                                                                                                                                                                                                                                                                                                                                                                                | (3M)<br>(CO4,L3,L2)  |              |            |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |                             |
| 4C      | Suppose that a disk drive has 5000 cylinders numbered 0 to 4999. The drive is currently serving a request at cylinder 143 and the previous request was at cylinder 125. The queue of pending requests in FIFO order is 86, 1470, 913, 1774, 948, 1022, 1750, 130. Starting from the current head position, what is the total distance in cylinders that the disk arm moves to satisfy all the pending requests using Shortest Seek Time First (SSTF) algorithm.                                                                                                                                                                                        | (2M)<br>(CO5,L3)     |              |            |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |                             |
| 5A      | Discuss the strengths and weaknesses of implementing an access matrix using access lists that are associated with objects. Illustrate how access matrix is used for different operations.                                                                                                                                                                                                                                                                                                                                                                                                                                                              | (5M)<br>(CO5,L2)     |              |            |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |                             |
| 5B      | Write a program for file locking using Java API. Explain in detail about file mounting with suitable example.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | (3M)<br>(CO5,L4)     |              |            |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |                             |
| 5C      | Explain Acyclic Graph Directory Structure and Tree level directory structure.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | (2M ,L2)<br>(CO5,L2) |              |            |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |    |   |   |                             |



# Question Paper

Exam Date & Time: 28-Jun-2024 (02:30 PM - 05:30 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

Department of Information and Communication Technology  
IV Semester B.Tech(CCE) Degree Make-up Examinations June 2024

### OPERATING SYSTEMS [ICT 2227]

Marks: 50

Duration: 180 mins.

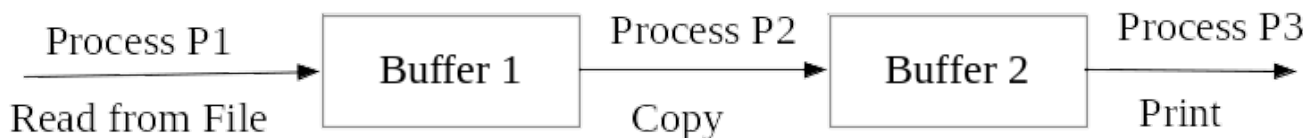
A

Answer all the questions.

Section Duration: 180 mins

Missing Data suitably be assumed

- 1) Consider three processes (P1, P2, and P3) that are involved in printing a file (as shown below). Process P1 reads the data from the disc into Buffer 1, then Process P2 copies the data from Buffer 1 to Buffer 2, and finally Process P3 prints the data from Buffer 2. Assume all three processes operate on one file at a time, and buffers 1 and 2 have a capacity of 1 file/record. Write pseudo code as a solution that uses semaphores to coordinate the three processes. Represent the operations executed in the critical section, using proper comment lines. (5)



- 2) Write pseudo code to demonstrate how Dekker's algorithm can be used as a solution to the critical section problem for two processes P0 and P1. (3)
- 3) With a neat diagram, illustrate and explain how operating system manages the ready queue while serving system services like handling I/O requests and interrupts. (2)
- 4) The memory access time is 1 nanosecond for a read operation with a hit in TLB, 5 nanoseconds for a read operation with a miss in TLB, 2 nanoseconds for a write operation with a hit in TLB and 10 nanoseconds for a write operation with a miss in TLB. Execution of a sequence of instructions involves 100 instruction fetch operations, 60 memory read operations and 40 memory write operations. The TLB hit ratio is 90%. Determine the average memory access time (in nanoseconds) in executing the sequence of all instructions by providing detailed steps. (5)
- 5) Consider processes  $P=\{S1, S2, S3\}$  and resources  $R=\{R1, R2, R3\}$  with one instance of resource type R1, two instances of resource type R2 and one instance of resource type R3. Currently S1 is holding an instance of R2 and waiting for instance of R1. S2 is holding an instance of resource type R1, one instance of resource type R2 and waiting for an instance of resource type of R3. S3 is holding an instance of resource type R3. (3)
- (a) Draw Resource Allocation Graph
- (b) Check is there any deadlock. Justify your answer.
- (c) If there is a deadlock, write how to make it deadlock free. If not, write all possible safe sequences.
- 6) Consider the set of processes burst time in milliseconds as given below. Find the waiting time, response time and turnaround time of each process using round robin scheduling with time quantum 3ms. (2)

| Process | Arrival Time | Burst Time |
|---------|--------------|------------|
| P1      | 3            | 3          |
| P2      | 2            | 5          |

|    |   |    |
|----|---|----|
| P3 | 1 | 8  |
| P4 | 0 | 10 |

- 7) Consider a swapping system in which memory consists of the following fragment sizes in order: 10K, 4K, 20K, 18K, 7K, 9K, 12K, and 15K. Which fragment is taken for successive segment requests of: a) 12K b) 10K c) 9K for first fit, best fit, worst fit memory allocation algorithms? Which memory allocation algorithm is a better solution for applications that need very large process sizes? Justify your answer. (5)

- 8) Consider the following snapshot of a system: (3)

|    | Allocation |   |   |   | Max |   |   |   | Available |   |   |   |
|----|------------|---|---|---|-----|---|---|---|-----------|---|---|---|
|    | A          | B | C | D | A   | B | C | D | A         | B | C | D |
| P0 | 0          | 0 | 1 | 2 | 0   | 0 | 1 | 2 | 1         | 5 | 2 | 0 |
| P1 | 1          | 0 | 0 | 0 | 1   | 7 | 5 | 0 |           |   |   |   |
| P2 | 1          | 3 | 5 | 4 | 2   | 3 | 5 | 6 |           |   |   |   |
| P3 | 0          | 6 | 3 | 2 | 0   | 6 | 5 | 2 |           |   |   |   |
| P4 | 0          | 0 | 1 | 4 | 0   | 6 | 5 | 6 |           |   |   |   |

Use Banker's algorithm and answer the following:

- Find need matrix.
  - Is the system in a safe state? If yes, write the safe sequence, show steps.
  - If a request from process P1 arrives for (0,4,2,0), can the request be granted immediately? If yes, show the steps.
- 9) With a neat diagram explain the working of medium-term scheduler. (2)
- 10) Compare and contrast three methods of managing free disk space: Linked List (Free List), Grouping, and Counting. Discuss their advantages and disadvantages in managing free disk space with an example for each. (5)
- 11) With the help of neat diagram show the stack implementation of LRU page replacement algorithm for the memory references: 0300, 0632, 0101, 0812, 0302, 0303, 0304, 0301, 0811, 0302, 0303, 0304, 0301, 0810, 0302, 0303, 0304, 0301, 0809, 0302, 0305. Find the reference string if there exist 100 bytes per page, calculate the number of page faults and page hits, show frame allocation. (3)
- 12) Differentiate between the following frame allocation strategies of virtual memory. (2)
- Equal allocation and Proportional allocation
  - Global allocation and Local Allocation
- 13) Suppose a disk drive has 5000 cylinders, numbered 0 to 4999. Consider a disk queue with requests for i/o to blocks on cylinder: 106, 1370, 814, 1671, 863, 1416, 1149, 1620, 140. Assume that disk head is currently at cylinder 122. With neat diagram show the trace of SSTF, SCAN, and C-SCAN, disk scheduling algorithm. Assume the direction of movement is towards the higher cylinders in case of SCAN and C-SCAN. Calculate the head movement and the average seek time for each of these disk scheduling algorithms (5)

- 14) (3)

Given a computer system with a memory access time of 150 ns and a page fault rate of 0.08 (or 8%), calculate the effective memory access time if the page fault service time is 25 ms. Explain the significance of this effective memory access time value in terms of system performance. How would you optimize the system to reduce the effective memory access time and improve overall performance?

- 15) Consider a real time system having three periodic tasks with their periods and execution times as  $T_i(p_i, e_i)$  :  $T_1(20,3)$  (2)  
 $T_2(5,2)$ ,  $T_3(10,2)$ . Check is it RMA schedulable? Draw the pre-emptive RMA (Rate Monotonic Analysis) schedule of jobs on single processor for timeline from 0-20units.

-----End-----

# Question Paper

Exam Date & Time: 11-May-2024 (02:30 PM - 05:30 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

Department of Information and Communication Technology  
IV semester B.Tech(CCE) End Semester Examinations May 2024

### OPERATING SYSTEMS [ICT 2227]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

Missing data if any can be assumed with proper reasoning.

- 1a) Suppose there are five processes P1, P2, P3, P4 and P5 that need to access a shared resource R. (5)  
The access policy for R is as follows: P1 and P2 can access R concurrently, but no other process can access R when they are accessing it. P3, P4 and P5 can access R concurrently, but no other process can access R when they are accessing it. Design a solution using semaphores to implement this policy and check for possibility of deadlock. Demonstrate using a pseudocode for each process.
- 1b) Consider a scenario where two processes P1 and P2, are trying to update a shared bank account balance. Design a solution to the critical section problem and demonstrate correct implementation of deposit and withdraw operations on the shared bank account, balance initially set to 1000. (3)
- 1c) With a neat diagram, illustrate how transition of operating system from user mode to kernel mode happens when an application uses a system call to create a process. (2)
- 2a) Consider a machine with 64 MB byte addressable physical memory and a 32bit virtual address space. If the page size is 4 KB, find the following with proper justification. (5)
- (a) Number of bits in physical address
  - (b) Number of frames in main memory
  - (c) Number of bits in frame number and page offset. What is the approximate size of the page table?
  - (d) Process size
  - (e) Number of entries in page table
- 2b) A system has 12 resources of A type, 14 resources of B type and 15 resources of C type. Find the current availability of the resources and apply Banker's algorithm and check whether the system is in safe state. Write the safe sequence if the system is deadlock free. Show the steps. (3)

| Process | MAX |   |   | Allocation |   |   |
|---------|-----|---|---|------------|---|---|
|         | A   | B | C | A          | B | C |
| P1      | 8   | 7 | 5 | 2          | 3 | 4 |
| P2      | 7   | 4 | 4 | 4          | 2 | 3 |
| P3      | 2   | 5 | 5 | 1          | 4 | 4 |
| P4      | 5   | 5 | 4 | 3          | 3 | 4 |

- 2c) Consider the set of processes and their priority (smaller the number, higher the priority), arrival time (2) and their burst time in milliseconds as given. Schedule the set of processes using pre-emptive priority scheduling and first come first serve algorithms. Draw Gantt charts for both.

| Process | Arrival Time | Burst Time | Priority |
|---------|--------------|------------|----------|
| P1      | 1            | 6          | 1        |
| P2      | 0            | 5          | 2        |
| P3      | 2            | 2          | 1        |
| P4      | 3            | 3          | 2        |

- 3a) Consider six memory partitions of size 200 KB, 400 KB, 600 KB, 500 KB, 300 KB and 250 KB in (5) order. These partitions need to be allocated to four processes of sizes 357 KB, 210 KB, 468 KB and 491 KB in that order. Show memory allocation using first fit, best fit and worst fit algorithms. Calculate internal and external fragmentation if any in each case.

- 3b) Consider a system employing a deadlock-avoidance scheme for single instance of each resource (3) type. Let P1, P2 and P3 be three processes in the system. Let P1 declare that it may request resources W, X and Y; P2 declare that it may request resources X, Y and Z; P3 declare that it may request resources W, Y and Z. Construct the resource-allocation graph by considering the resource requests declared by P1, P2 and P3. Consider each of the following pairs of independent resource requests to determine whether requests can be granted without a deadlock possibility. Justify your answer.

- P1 requesting X and P2 requesting Z
- P2 requesting Y and P3 requesting W
- P1 requesting Y and P3 requesting W
- P1 requesting X and P2 requesting Y

- 3c) What is the advantage and dis-advantage of Shortest Job First scheduling algorithm? Consider a (2) system of 5 processes and their burst times P1(10), P2(5), P3(3), P4(4) and P5(5). Predict the burst time of P6, using exponential averaging method, if  $\tau=10\text{ms}$  and  $\alpha=0.5$ .

- 4a) Given main memory address sequence which are referred by the user: 1536, 2912, 4096, 6144, (5) 7680, 1536, 2912, 4096, 6144, 7680, 1536, 2912, 4096, 6144, 7680, 1536. If there exists 100 bytes per page, find the reference string. Apply the following page replacement algorithms and find the number of page faults if the main memory has 4 frames.

- First In First Out (FIFO). Also, check if there exists Belady's anomaly or not.
- Least Recently Used page replacement algorithm (LRU)
- Optimal page replacement algorithm

- 4b) In a computer system utilizing demand paging, the system experiences page faults based on a (3)

page fault rate 'p', where  $0 \leq p \leq 1.0$ . The system has the following characteristics:

Memory Access Time: 330 nanoseconds

Page fault overhead: 5 milliseconds.

Swap page out time: 20 milliseconds.

Swap page in time: 25 milliseconds.

Restart overhead: 10 milliseconds.

Calculate the Effective Access Time (EAT) for the following scenarios:

Case 1: When there are no page faults ( $p=0$ )

Case 2: When there is a 50% chance of a page fault ( $p=0.5$ )

Case 3: When every memory reference results in a page fault ( $p=1.0$ )

- 4c) A Unix file system uses a i-node structure containing 10 direct block addresses, one indirect, one double and one triple indirect block addresses. The size of each block is 128bytes and the size of each block address is 8bytes. Find the maximum possible file size. (2)
- 5a) Consider a scenario if there exist three files: File A, File B, and File C. File A takes 3 blocks in the disk, File B takes 2 blocks in the disk, File C takes 5 blocks in the disk. With neat diagrams/table show how these three files (File A, File B, and File C) are allocated to the disk using the following disk allocation methods. (5)
- a. Contiguous allocation
  - b. Linked Allocation
  - c. Indexed Allocation
- 5b) Suppose the total physical memory frames (m) is 300, Number of processes (n) available in the system is 5, Memory requirements for each process: (3)
- Process 1: 40 frames, Process 2: 30 frames, Process 3: 20 frames, Process 4: 50 frames and Process 5: 60 frames.
- a. Calculate the total number of frames allocated to each process using Equal allocation and Proportional allocation strategy.
  - b. Calculate Internal Fragmentation and External Fragmentation using Equal allocation and Proportional allocation strategy.
- 5c) Consider a real time system having three periodic tasks with their period, execution time and deadlines  $T_i(p_i, e_i, d_i)$  as  $T_1(10,1,6)$ ,  $T_2(5,1,4)$ ,  $T_3(3,1,2)$ . Draw the pre-emptive Earliest Deadline First schedule of jobs on a single processor for timeline from 0-10 time units (2)

-----End-----



# Question Paper

Exam Date & Time: 01-Jul-2023 (02:30 PM - 05:30 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

FOURTH SEMESTER B.TECH. (INFORMATION TECHNOLOGY) DEGREE EXAMINATIONS - JUNE/JULY 2023  
SUBJECT: ICT 2258/IT-2258 OPERATING SYSTEMS  
(MAKEUP)

Marks: 50

Duration: 180 mins.

Answer all the questions.

Missing data, if any, may be suitably assumed.

- 1A) Suppose that a disk drive has 5000 cylinders, numbered 0 to 4999. The drive is currently serving a request at cylinder 143, and the previous request was at cylinder 125. The queue of pending requests, in FIFO order, is 86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130. Starting from the current head position, what is the total distance ((in cylinders) that the disk arm moves to satisfy all the pending requests, for each of the following disk scheduling (5)
- (a) SCAN (b) LOOK (c) C-SCAN

- 1B) Consider a file currently consisting of 100 blocks. Assume that the filecontrol block (and the index block, in the case of indexed allocation) is already in memory. Calculate how many disk I/O operations are required for contiguous, linked, and indexed (single-level) allocation strategies, if, for one block, the following conditions hold. In the contiguous-allocation case, assume that there is no room to grow at the beginning but there is room to grow at the end. Also assume that the block information to be added is stored in memory. (3)
- (i) The block is added at the beginning.  
(ii) The block is added in the middle.  
(iii) The block is added at the end.

- 1C) Suppose that the following processes arrive for execution at the times indicated. Each process will run the listed amount of time. In answering the questions, use nonpreemptive scheduling and base all decisions on the information you have at the time the decision must be made. (2)

| PROCESS | ARRIVAL TIME | BURST TIME |
|---------|--------------|------------|
| P1      | 0.0          | 8          |
| P2      | 0.4          | 4          |
| P3      | 1.0          | 1          |

- i) What is the average turnaround time for these processes with the FCFS scheduling algorithm?  
ii) What is the average turnaround time for these processes with the SJF scheduling algorithm?

- 2A) A system has a physical memory of 128 KB and uses a paging scheme with a page size of 2 KB. If the system uses an inverted page table and each entry in the table requires 16 bytes, what is the maximum number of page table entries that can be accommodated? (5)

- 2B) Consider a file system with a maximum file size of 1 GB and an i-node size of 256 bytes. If each i-node can store 12 direct block pointers, a single indirect block pointer, and a double indirect block pointer, what is the maximum number of blocks that a file can address directly? (3)

- 2C) Illustrates the steps used for translating logical address into physical address with suitable example and necessary diagram. (2)

- 3A) Assume that you have the following attributes for a virtual memory system. (5)

i) Three frames in Main Memory  
 ii) Two processes p1 and p2  
 iii) Number of pages in Virtual memory are 6(A,B,C,D,E,F).  
 The page reference string encoded below has < process name-page name> For example < p1-A> means p1 process A page.  
 Page reference String: < p1-A>,< p2-B>,< p2-C>,< p1-B>,< p2-A>,< p1-B>,< p2-D>,< p2-A>,< p1-B>,< p2-E>,< p2-F>,< p1-C>.

Show the frame contents and calculate page-fault rate for each of the following page replacement policies.

i) LRU (Global page replacement)  
 ii) FIFO (Local page replacement) Assume that process p1 is assigned 2 pages and p2 is assigned 1 page

3B) To transfer a 4KB block on a 5400 RPM disk with a 4ms average seek time, 1Gb/sec transfer rate (3)  
 with a 0.2ms controller overhead, what is the total average access time taken?  
 Hint: 1GB=1048576KB.

3C) Consider a logical address space of 64 pages of 1KB each, mapped onto a physical memory of 32 (2)  
 frames.  
 i) How many bits are there in the logical address?  
 ii) How many bits are there in the physical address?

4A) Consider a system with five processes P1 through P5 and three resource types x,y,z. Resource (5)  
 type x has 10 instances, y has 5 instances and type z has 12 instances. Suppose at time  $t_0$   
 following snapshot of the system are given in Table 4A.

Table 4A

| Process | Allocation |   |   | Max |   |   |
|---------|------------|---|---|-----|---|---|
|         | X          | Y | Z | x   | y | Z |
| P1      | 2          | 0 | 2 | 3   | 1 | 3 |
| P2      | 1          | 0 | 3 | 2   | 0 | 4 |
| P3      | 0          | 2 | 2 | 4   | 3 | 3 |
| P4      | 4          | 3 | 3 | 5   | 3 | 3 |
| P5      | 0          | 0 | 2 | 2   | 2 | 2 |

i) Check whether the system is in safe state. If so, obtain the safe sequence.  
 ii) Whether the additional requests for x and y from P1 can be granted or not.  
 iii) If process P3 requests (2,1,1), after step ii, can it be granted? If yes, obtain the safe sequence.

4B) Differentiate between system calls and function calls. Illustrate dual mode operation by showing the (3)  
 standard C library's system call.

4C) A file system uses i-node structure which contains 8 direct block address, one indirect block, one (2)  
 double indirect and one triple indirect block address. The size of the disk block is 128bytes. The size  
 of each block address is 8bytes. Find the maximum possible file size.

5A) Code Q5A represents assembly language code for a pair of processes that share the variable (5)  
 result. The initial value of the *result* before execution of either process begins is 10. Here, *LD* loads  
 the memory variable into the target register, the *ADD* instruction adds values, and *ST* stores the  
 register value into the memory variable. If processes X and Y are run on a timesharing system,  
 there are six possible orders in which the LD and ST instructions might be executed. For each of  
 the orders, the final value of the counter variable could be 11, 12, or 13. i) Add semaphores (with  
 initial values) so that the final value of the counter is 12. ii) Add semaphores (with initial values), so  
 that the final value of the counter is not 13. iii) Add semaphores (with initial values), so that process  
 X executes only after the execution of process 2.

|                                                                             |                                                                            |
|-----------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>Process X:</b><br>X1: LD(result,R0)<br>ADD (R0, 1)<br>X2: ST(R0, result) | <b>Process Y:</b><br>Y1: LD(result,R0)<br>ADD(R0, 2)<br>Y2: ST(R0, result) |
|-----------------------------------------------------------------------------|----------------------------------------------------------------------------|

Code Q5A.

5B) What type of OS is used in embedded system? Give its characteristics. (3)

5C) With two examples, demonstrate two multithreading issues as compared to single-threaded applications. (2)

-----End-----

# Question Paper

Exam Date & Time: 01-Jul-2023 (02:30 PM - 05:30 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

FOURTH SEMESTER B.TECH. (INFORMATION TECHNOLOGY) DEGREE EXAMINATIONS - JUNE/JULY 2023  
SUBJECT: ICT 2258/IT-2258 OPERATING SYSTEMS  
(MAKEUP)

Marks: 50

Duration: 180 mins.

Answer all the questions.

Missing data, if any, may be suitably assumed.

- 1A) Suppose that a disk drive has 5000 cylinders, numbered 0 to 4999. The drive is currently serving a request at cylinder 143, and the previous request was at cylinder 125. The queue of pending requests, in FIFO order, is 86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130. Starting from the current head position, what is the total distance ((in cylinders) that the disk arm moves to satisfy all the pending requests, for each of the following disk scheduling (5)
- (a) SCAN (b) LOOK (c) C-SCAN

- 1B) Consider a file currently consisting of 100 blocks. Assume that the filecontrol block (and the index block, in the case of indexed allocation) is already in memory. Calculate how many disk I/O operations are required for contiguous, linked, and indexed (single-level) allocation strategies, if, for one block, the following conditions hold. In the contiguous-allocation case, assume that there is no room to grow at the beginning but there is room to grow at the end. Also assume that the block information to be added is stored in memory. (3)
- (i) The block is added at the beginning.  
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- 1C) Suppose that the following processes arrive for execution at the times indicated. Each process will run the listed amount of time. In answering the questions, use nonpreemptive scheduling and base all decisions on the information you have at the time the decision must be made. (2)

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- 2A) A system has a physical memory of 128 KB and uses a paging scheme with a page size of 2 KB. If the system uses an inverted page table and each entry in the table requires 16 bytes, what is the maximum number of page table entries that can be accommodated? (5)

- 2B) Consider a file system with a maximum file size of 1 GB and an i-node size of 256 bytes. If each i-node can store 12 direct block pointers, a single indirect block pointer, and a double indirect block pointer, what is the maximum number of blocks that a file can address directly? (3)

- 2C) Illustrates the steps used for translating logical address into physical address with suitable example and necessary diagram. (2)

- 3A) Assume that you have the following attributes for a virtual memory system. (5)

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|         | X          | Y | Z | x   | y | Z |
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| P2      | 1          | 0 | 3 | 2   | 0 | 4 |
| P3      | 0          | 2 | 2 | 4   | 3 | 3 |
| P4      | 4          | 3 | 3 | 5   | 3 | 3 |
| P5      | 0          | 0 | 2 | 2   | 2 | 2 |

i) Check whether the system is in safe state. If so, obtain the safe sequence.  
 ii) Whether the additional requests for x and y from P1 can be granted or not.  
 iii) If process P3 requests (2,1,1), after step ii, can it be granted? If yes, obtain the safe sequence.

4B) Differentiate between system calls and function calls. Illustrate dual mode operation by showing the (3)  
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|                                                                             |                                                                            |
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Code Q5A.

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-----End-----

# Question Paper

Exam Date & Time: 01-Jul-2023 (02:30 PM - 05:30 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

**FOURTH SEMESTER B.TECH. (INFORMATION TECHNOLOGY) DEGREE EXAMINATIONS - JUNE/JULY 2023**  
**SUBJECT: ICT 2258/IT-2258 OPERATING SYSTEMS**  
**(MAKEUP)**

**Marks: 50**

**Duration: 180 mins.**

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2B) Consider a file system with a maximum file size of 1 GB and an i-node size of 256 bytes. If each i-node can store 12 direct block pointers, a single indirect block pointer, and a double indirect block pointer, what is the maximum number of blocks that a file can address directly? (3)

2C) Illustrates the steps used for translating logical address into physical address with suitable example and necessary diagram. (2)

3A) Assume that you have the following attributes for a virtual memory system. (5)

i) Three frames in Main Memory  
 ii) Two processes p1 and p2  
 iii) Number of pages in Virtual memory are 6(A,B,C,D,E,F).  
 The page reference string encoded below has < process name-page name> For example < p1-A> means p1 process A page.  
 Page reference String: < p1-A>,< p2-B>,< p2-C>,< p1-B>,< p2-A>,< p1-B>,< p2-D>,< p2-A>,< p1-B>,< p2-E>,< p2-F>,< p1-C>.

Show the frame contents and calculate page-fault rate for each of the following page replacement policies.

i) LRU (Global page replacement)  
 ii) FIFO (Local page replacement) Assume that process p1 is assigned 2 pages and p2 is assigned 1 page

3B) To transfer a 4KB block on a 5400 RPM disk with a 4ms average seek time, 1Gb/sec transfer rate (3)  
 with a 0.2ms controller overhead, what is the total average access time taken?  
 Hint: 1GB=1048576KB.

3C) Consider a logical address space of 64 pages of 1KB each, mapped onto a physical memory of 32 (2)  
 frames.  
 i) How many bits are there in the logical address?  
 ii) How many bits are there in the physical address?

4A) Consider a system with five processes P1 through P5 and three resource types x,y,z. Resource (5)  
 type x has 10 instances, y has 5 instances and type z has 12 instances. Suppose at time  $t_0$   
 following snapshot of the system are given in Table 4A.

Table 4A

| Process | Allocation |   |   | Max |   |   |
|---------|------------|---|---|-----|---|---|
|         | X          | Y | Z | x   | y | Z |
| P1      | 2          | 0 | 2 | 3   | 1 | 3 |
| P2      | 1          | 0 | 3 | 2   | 0 | 4 |
| P3      | 0          | 2 | 2 | 4   | 3 | 3 |
| P4      | 4          | 3 | 3 | 5   | 3 | 3 |
| P5      | 0          | 0 | 2 | 2   | 2 | 2 |

i) Check whether the system is in safe state. If so, obtain the safe sequence.  
 ii) Whether the additional requests for x and y from P1 can be granted or not.  
 iii) If process P3 requests (2,1,1), after step ii, can it be granted? If yes, obtain the safe sequence.

4B) Differentiate between system calls and function calls. Illustrate dual mode operation by showing the (3)  
 standard C library's system call.

4C) A file system uses i-node structure which contains 8 direct block address, one indirect block, one (2)  
 double indirect and one triple indirect block address. The size of the disk block is 128bytes. The size  
 of each block address is 8bytes. Find the maximum possible file size.

5A) Code Q5A represents assembly language code for a pair of processes that share the variable (5)  
 result. The initial value of the *result* before execution of either process begins is 10. Here, *LD* loads  
 the memory variable into the target register, the *ADD* instruction adds values, and *ST* stores the  
 register value into the memory variable. If processes X and Y are run on a timesharing system,  
 there are six possible orders in which the LD and ST instructions might be executed. For each of  
 the orders, the final value of the counter variable could be 11, 12, or 13. i) Add semaphores (with  
 initial values) so that the final value of the counter is 12. ii) Add semaphores (with initial values), so  
 that the final value of the counter is not 13. iii) Add semaphores (with initial values), so that process  
 X executes only after the execution of process 2.

|                                                                             |                                                                            |
|-----------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>Process X:</b><br>X1: LD(result,R0)<br>ADD (R0, 1)<br>X2: ST(R0, result) | <b>Process Y:</b><br>Y1: LD(result,R0)<br>ADD(R0, 2)<br>Y2: ST(R0, result) |
|-----------------------------------------------------------------------------|----------------------------------------------------------------------------|

Code Q5A.

5B) What type of OS is used in embedded system? Give its characteristics. (3)



5C) With two examples, demonstrate two multithreading issues as compared to single-threaded applications. (2)

-----End-----